

**AMENDMENT NO. 1 JANUARY 1998
TO
IS 101(PART 5/SEC 2) : 1988 METHODS OF SAMPLING
AND TEST FOR PAINTS, VARNISHES AND RELATED
PRODUCTS**

PART 5 MECHANICAL TESTS

Section 2 Flexibility and Adhesion

(Third Revision)

(Page 1, clause 2.2.2, line 3) — Substitute one of the diameters of the mandrels '6.25 mm' for '6 mm'.

(Page 2, clause 2.2.3, line 4) — Substitute one of the diameters of the mandrels '6.25 mm' for '6 mm'.

(Page 5, clause 3.2.3) — Substitute the following for the existing clause:

'3.2.3 Needle — The scratching needle has a hard hemispherical tip of 1 mm of diameter. The hemispherical tip shall be firmly attached and the exposed part shall be free from any contaminants.

NOTE — Details about the tip and the needle are given in Annex A for guidance.'

(Page 5, clause 3.3) — Substitute the following for the existing clause:

'3.3 Test Panels — Unless agreed for, the panels shall be of tin plate, burnishing steel or hard aluminium prepared by acid chromating.'

(Page 12, clause 5.6.2) — Insert the following Annex at the end:

**ANNEX A
(Clause 3.2.3)**

**PROCEDURE FOUND TO BE CONVENIENT FOR MANUFACTURE
AND RE-TIPPING OF SCRATCH TEST NEEDLES**

A-1 Manufacture of New Needles

A-1.1 For a batch of shanks in the vertical position with the dimpled end uppermost by inserting them in a tray made from perforated sheet metal.

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A-1.2 Place a very small quantity of a suitable solder paste on the end of each shank, and then put a steel ball in position, relying on the solder paste to hold the ball in position at this stage.

NOTE — The amount of solder paste to be used is judged by experience, insufficient will not make a firm soldered joint and too much will lead to more or less complete envelopment of the steel ball.

A-1.3 Place the tray of shanks in an oven or muffle furnace adjusted to a temperature of 210°C to 220°C for about 5 min to melt the solder and thus secure the ball in the dimple at the end of the shank.

A-1.4 Remove the needles from the source of heat, allow to cool, and wipe each ball to remove flux residues.

A-1.5 Check that the ball is securely fixed and that there is no solder on the part of the ball which will be used to make the scratch.

A-2 Re-tipping of Needles

A-2.1 Fix the needles in a tray made from perforated sheet metal as described in A-1.1 above.

A-2.2 Place the tray of needles in an oven or muffle furnace adjusted to a temperature of 210°C to 220°C and, as soon as the solder has softened, remove the tray and brush the end of the needles with a clean brush in order to dislodge the steel balls.

A-2.3 When the shanks have cooled, proceed as described in A-1.2 to A-1.5.

A-3 Guidance on Tip Material

The following tips are in common use:

- | | |
|---------------------------------|---|
| a) Steel Balls | Depending on the circumstances, this type of tip may show signs of early wear. In view of the ease with which needles with steel balls can be made, it is recommended that they be used once only and then re-tipped. |
| b) Tungsten Carbide Tips | These tips last much longer than steel tips and can be obtained commercially. |

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c) Ruby Tips

These (ceramic) tips last a very long time. The tips have to be attached to the shank by glueing rather than by soldering and can be obtained commercially.

In all cases, it is important that a suitable steel shank is used which will hold the tip securely in the correct position. An example of a suitable shank is given in Fig. 6 of this standard.

(CHD 020)